Camille Vedani

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Amanda Henriksson, Erasmus+ Mobility Coordinator
International Office
Luleå University of Technology
SE-971 87 Luleå, Sweden

Re: Application for a six-month study period in embedded and cyber-physical systems

Dear Amanda Henriksson,

I am a second-year engineering student at ENSEEIHT (Toulouse INP), in the Computer Science & Networks apprenticeship track (FISA), and an apprentice software engineer at Kyber, a French open-source start-up building infrastructure for real-time interactions. At Kyber, I develop an SDK that enables remote control of drones, robots and other machines with less than 20 ms of latency: I implement network protocols based on QUIC and WebTransport and optimise our Rust/WebAssembly codebase for browser integration. My path is deliberately broad: two years of preparatory classes in life sciences (CPGE BCPST), a double bachelor's degree in mathematics and computer science at Aix-Marseille University, and a first master's year in embedded and connected systems at Toulouse III. Two research internships, on state reduction for probabilistic automata at LIS (CNRS) and on execution decision diagrams for worst-case execution time analysis at IRIT, taught me the rigour of academic research. I also designed the control logic of an autonomous sailboat model, which taught me how software behaves once it meets real sensors and actuators. All of this serves one goal: designing safe, open medical embedded devices such as insulin pumps.

Luleå University of Technology attracts me for its applied, industry-oriented research and teaching in embedded and cyber-physical systems, in particular within the Embedded Intelligent Systems Lab (EISLAB), and for its unique expertise in space engineering, close to the Esrange Space Center. As a long-time enthusiast of rocket launches and aviation, I find the idea of studying where embedded systems must operate in the most demanding conditions especially motivating. LTU's human scale and its strong links with Swedish industry also promise an environment where students quickly take on real responsibilities.

I would like to spend one semester (six months) at your university, within the exchange agreement between ENSEEIHT (Toulouse INP) and LTU. I would bring my skills in Rust, C/C++, Python, ROS and embedded Linux, my experience of low-latency networking for remote-controlled systems, and the autonomy I have developed while working fully remotely in an international, English-speaking team. In return, this semester would let me gain first-hand experience of robust embedded systems designed for harsh environments, a key requirement for medical devices too, while fulfilling the international mobility required by my engineering degree. I would be glad to provide any further information about my academic record. My CV and portfolio are attached.

Thank you for considering my application. I look forward to hearing from you.

Sincerely,

Camille Vedani